

Suho Shin

Ph.D. Student at University of Maryland, College Park

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RESEARCH INTEREST

I am interested in mechanism design and market design, broadly construed. A core thread throughout my research is the development of the foundations of delegated decision-making under uncertainty, with various applications spanning from classic markets and organization, modern online platforms, to emerging ecosystem shaped by generative AI.

EMPLOYMENT

Stanford University

Motwani Postdoctoral Fellow

California, USA

Aug. 2026 – Aug. 2027

- Host: [Prof. Amin Saberi](#)

Massachusetts Institute of Technology

Postdoctoral Scholar

Massachusetts, USA

Sep. 2027 – Aug. 2028 (Expected)

- Host: [Prof. Negin Golrezaei](#)

EDUCATION

University of Maryland, College Park

Ph.D. in Computer Science

Maryland, USA

Sep. 2022 – May. 2026

- Advisor: [Prof. MohammadTaghi Hajiaghayi](#)

Korea Advanced Institute of Science and Technology

M.S. in Electrical Engineering

Daejeon, South Korea

Mar. 2016 – Jan. 2018

- Advisor: [Prof. Yung Yi](#)

Korea Advanced Institute of Science and Technology

B.S. in Mathematical Sciences

Daejeon, South Korea

Mar. 2011 – Feb. 2016

PUBLICATION

(α, β) denotes alphabetical order of authorship.

Optimally Curating an Event

α, β M. Hajiaghayi, S. Lahaie, M. Mahdavi, S. Shin

IJCAI'26

Optimal Contest for Recommender Systems

α, β N. Golrezaei, M. Hajiaghayi, S. Shin

STOC'26

- 2025 Market Innovation Workshop

- 2025 PSOR best video competition, finalists ([Youtube link](#))

Single-Sample Bilateral Trade with a Broker

WWW'26

α, β M. Hajiaghayi, G. Peng, S. Shin

Algorithmic Delegated Choice: An Annotated Reading List

SIGecom Exchanges Vol 23.1, 2025

α, β M. Hajiaghayi, S. Shin

Delegation with Costly Inspection

EC'25

α, β M. Hajiaghayi, P. Krysta, M. Mahdavi, S. Shin

Delegated Choice with Combinatorial Constraints

EC'25

α, β K. Banihashem, M. Hajiaghayi, P. Krysta, S. Shin

- Under review at *Operations Research*

Tokenized Bandit for LLM Decoding and Alignment

ICML'25

S. Shin, C. Yang, H. Xu, M. Hajiaghayi

- EC'25 workshop on Info/Econ/LLMs

- EC'25 workshop on Human-AI collab

Replication-proof Bandit Mechanism Design with Bayesian Agents

AAAI'25 (oral)

S. Shin, S. Esmaili, M. Hajiaghayi

Robust and Performance Incentivizing Algorithms for Bandits with Strategic Agents

AAAI'25

S. Esmaili, S. Shin, A. Slivkins

Gains-from-Trade in Bilateral Trade with a Broker

SODA'25

α, β I. Hajiaghayi, M. Hajiaghayi, G. Peng, S. Shin

Online Advertisements with LLMs

SIGecom Exchanges Vol 22.2, 2025

α, β S. Feizi, M. Hajiaghayi, K. Rezaei, S. Shin

Ad Auctions for LLMs via Retrieval Augmented Generation

NeurIPS'24

α, β M. Hajiaghayi, S. Lahaie, K. Rezaei, S. Shin

- EC'24 Workshop on Frontiers of Online Advertising: Autobidding, GenAI, and Beyond

Dueling Over Dessert, Mastering the Art of Repeated Cake Cutting

NeurIPS'24

α, β S. Branzei, M. Hajiaghayi, R. Phillips, S. Shin, K. Wang

- Under review at *Math of Operations Research*

Fairness and Efficiency in Online Class Matching

NeurIPS'24

α, β M. Hajiaghayi, S. Jahan, M. Sharify, S. Shin, M. Springer

A Regret Analysis of Repeated Delegated Choice

AAAI'24

α, β M. Hajiaghayi, M. Mahdavi, K. Rezaei, S. Shin

An Improved Relaxation for Oracle-Efficient Adversarial Contextual Bandits

NeurIPS'23

α, β K. Banihashem, M. Hajiaghayi, S. Shin, M. Springer

Bandit Social Learning under Myopic Agents

NeurIPS'23

α, β K. Banihashem, M. Hajiaghayi, S. Shin, A. Slivkins

- Under review at *Math of Operations Research*
- *EC'24 Workshop on Information Acquisition*
- *AMLS'24 (best poster award)*

Delegating to Multiple Agents

EC'23

α, β M. Hajiaghayi, K. Rezaei, S. Shin

Multi-armed Bandit Algorithm against Strategic Replication.

AISTATS'22

S. Shin, S. Lee, J. Ok

Power of Bonus in Pricing for Crowdsourcing.

SIGMETRICS'22

S. Shin, H. Choi, Y. Yi, J. Ok

TEACHING

Analysis of Algorithms

Spring 2026

Teaching Assistant (~ 40 students)

UMD

Advanced Topics in Theory of Computing; Algorithmic Game Theory (CMSC858J)

Fall 2025

Guest Lecturer (4 full lectures)

UMD

Introduction to Computer Systems

Fall 2024

Teaching Assistant (~ 300 students)

UMD

Introduction to Computational Game Theory (CMSC474)

Fall 2023

Head Teaching Assistant (~ 100 students)

UMD

Design and Analysis of Computer Algorithms (CMSC451)

Spring 2023

Teaching Assistant (~ 100 students)

UMD

Introduction to Computational Game Theory (CMSC474)

Fall 2022

Head Teaching Assistant (~ 30 students)

UMD

Data Structure and Algorithms for Electrical Engineering (EE205)

Fall 2017

Teaching Assistant

KAIST

Computer Networks (EE323)

Spring 2017

Teaching Assistant

KAIST

HONOR & AWARD

METEOR Postdoctoral Fellowship (declined)

USA

Massachusetts Institute of Technology

May 2026

Gold Reviewer

N/A

International Conference on Machine Learning (ICML)

May 2026

Jane Street Graduate Research Fellowship (Honorable Mention)

USA

Jane Street

Feb 2026

PSOR Best Video Competition (Finalist)

USA

INFORMS Public and Societal Operations Research

Oct 2025

Outstanding Graduate Assistant Award AY 23-24*University of Maryland*

USA

Dec 2023

Graduate School Summer Research Fellowship (declined)*University of Maryland*

USA

Summer 2023

Dean's Fellowship*University of Maryland*

USA

Fall 2022 – Fall 2024

The National Scholarship for Science and Engineering*Korea Student Aid Foundation*

South Korea

Spring 2011 – Fall 2017

Dean's Award for Entrance*Korea Advanced Institute of Science and Technology*

South Korea

Spring 2011

Gold Medal, Korean Mathematical Olympiad*Korean Mathematical Society*

South Korea

Fall 2008

VISIT & INTERNSHIP

Visiting Student*Host: Prof. Haifeng Xu*

University of Chicago

Summer 2024

Visiting Student*Host: Prof. Piotr Krysta*

University of Liverpool

Jul. 2023

Visiting Student*Topic: Mathematics and Computer Science of Market and Mechanism Design*

SLMath, UC Berkeley

Jun. 2023

Research Intern*Host: Prof. Jungseul Ok*

Machine Learning Lab, Postech

Summer 2022

INDUSTRY EXPERIENCE

LINE plus Corporation, LINE Advertisement Platform*Data scientist, ML engineer*

Seongnam, South Korea

Oct. 2020 – Apr. 2022

Coupang, Product, Search and Discovery Platform*Software Engineer*

Jamsil, South Korea

Aug. 2018 – Sep. 2020

SERVICE

Program Committee*AIES'26, EC'26 EzPz Workshop, WINE'26, EC'26, AAAI'26, WINE'25, AAAI'25***Invited Reviewer***Algorithmica, NeurIPS'26, ICML'26, SODA'26, ICML'25, NeurIPS'24, ICML'24, NeurIPS'23, AISTATS'22*

MENTORING/ADVISING

Mentoring

Gary Peng

Dec. 2023 - Jun. 2026

- Undergrad at UMD, Now a Ph.D. at MIT EECS

- Received [CRA Outstanding Undergraduate Researcher Awards](#) ([link](#))

Mentoring

Aya Sghiouar (High school student at Bouskoura High School)

[HSRI'24](#)

Summer 2024

TALK

Optimal Contest beyond Convexity

KAIST ISE Winter Symposium

Daejeon, Korea

Dec. 2025

Modern Topics in Algorithmic Game Theory

Eat and Learn Seminar, Center for Algorithms & Optimization, Postech

Pohang, Korea

Dec. 2025

Tokenized Bandit for LLM Decoding and Alignment

Plutos AI

Virtual

Dec. 2025

Delegation with Costly Inspection

INFORMS Annual Meeting, Auctions and Market Design Award Session

Georgia, USA

Oct. 2025

Tokenized Bandit for LLM Decoding and Alignment

INFORMS Annual Meeting, Invited Session

Georgia, USA

Oct. 2025

Optimal Contest for Recommender Systems

INFORMS Annual Meeting, PSOR Business Meeting Award Session

Georgia, USA

Oct. 2025

Delegation with Costly Inspection

ACM Conference on Economics and Computation (EC)

Stanford, USA

Jul. 2025

Delegated Choice with Combinatorial Constraints

ACM Conference on Economics and Computation (EC)

Stanford, USA

Jul. 2025

Replication-proof Bandit Mechanism Design with Bayesian Agents

Association for the Advancement of Artificial Intelligence (AAAI)

Philadelphia, USA

Feb. 2025

Gains-from-Trade in Bilateral Trade with a Broker

Symposium on Discrete Algorithm (SODA)

New Orleans, USA

Jan. 2025

Prophet Inequality, Posted Pricing, and Delegated Choice

Microecon Theory Seminar, Sungkyunkwan University

Seoul, South Korea

Sep. 2024

Combinatorial Delegated Choice

East Asia Game Theory Conference

Jeju, South Korea

Aug. 2024

Ad Auctions for LLMs via Retrieval Augmented Generation

EC'24 Workshop on Frontiers of Online Advertising: Autobidding, GenAI, and Beyond

New Haven, USA

Jul. 2024

Delegated Choice, Prophet Inequality, and Beyond
Sigma Lab, UChicago

Chicago, USA
Jun. 2024

Topics in Economics and Computation
Machine Learning Lab, Postech

Pohang, South Korea
Jan. 2024

Delegating to Multiple Agents
ACM Conference on Economics and Computation (EC)

London, UK
Jul. 2023

Mechanism Design and Multi-armed Bandits
Machine Learning Lab, Postech

Virtual
Feb. 2022